[Ross-3] Section 4.3

State j is accessible from state i $\iff$ Starting from i, it is possible that the process ever enter j

$[\exists\, n \geq 1, \quad P_{ij}^{n} > 0]$ $\qquad$ $[(f_{ij}) > 0]$

Proof of $\Rightarrow$ is obvious

Proof of $\Leftarrow$: (to be shown)

$$f_{ij} > 0 \implies \exists\, n \geq 1 \quad P_{ij}^{n} > 0$$

suppose $f_{ij} > 0$ AND $\forall n \quad P_{ij}^{n} = 0$

$$f_{ij} = P\left[\text{ever enter } j \mid \text{starts in } i\right]$$

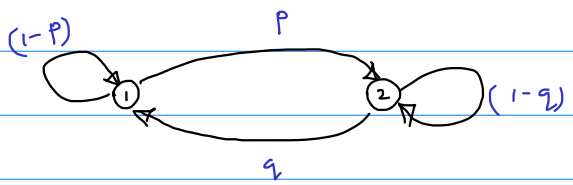
$$= P\left[\bigcup_{n=1}^{\infty} X_n = j \;\middle|\; X_0 = i\right]$$

$$\leq \sum_{n=1}^{\infty} P\left[X_n = j \mid X_0 = i\right]$$

$$= \sum_{n=1}^{\infty} P_{ij}^{n}$$

$$= 0 \quad \rightarrow \text{Contradiction}$$

Example [Kleinrock, Book, page 28]



$$f_1 = \sum_{i=1}^{\infty} f_1^{(i)}$$

$$= (1-p) + \sum_{n=2}^{\infty} pq(1-q)^{n-2}$$

$$= (1-p) + pq \sum_{n=0}^{\infty} (1-q)^{n}$$

check Ross-3 notations pg 189

$f_1^{(1)} = (1-p)$ $\qquad$ $P_{11}^{(1)} = (1-p)$

$f_1^{(2)} = pq$ $\qquad$ $P_{11}^{(2)} = (1-p)^2 + pq$

$f_1^{(3)} = p(1-q)(q)$ $\qquad$ $P_{11}^{(3)} = (1-p)^3 + (1-p)pq$

$f_1^{(4)} = p(1-q)^2 q$ $\qquad$ $P_{11}^{(4)} = (1-p)^4 + (1-p)^2 pq + p^2 q^2$

$f_1^{(5)} = p(1-q)^3 q$ $\qquad$ $P_{11}^{(5)} = (1-p)^5 +$ many combinations

$f_1^{(n)} = p(1-q)^{n-2} q$ $\qquad$ $n \geq 2$

recall $\sum_{k=1}^{n} ar^{k-1} = a\left(\frac{1-r^n}{1-r}\right) \Rightarrow \sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}$ $|r|<1$

$\sum_{n=1}^{\infty} p^n = \frac{p}{1-p}$; $\sum_{n=1}^{\infty} np^{n-1} = \frac{1}{(1-p)^2}$

Thus,

$$(1-p) + pq \sum_{n=0}^{\infty} (1-q)^n = (1-p) + pq \frac{1}{1-(1-q)} = (1-p) + \frac{pq}{q}$$

$$= 1-p+p = 1$$

* —— state (1) is a "recurrent" state. So is state (2)
  ↳ Compare with Theorem 1 Kleinrock page 29
* Moreover, state (1) is aperiodic because $\gamma = 1$
* Mean recurrent time of state 1 ?

$$M_1 = \sum_{n=1}^{\infty} n f_1^{(n)} = \sum_{n=2}^{\infty} n\, pq(1-q)^{n-2} + 1\cdot(1-p)$$

$$= (1-p) + (pq) \sum_{n=2}^{\infty} n r^{n-2} \quad ; \quad r = (1-q)$$

$$= (1-p) + \frac{pq}{r} \sum_{n=2}^{\infty} n r^{n-1}$$

$$= (1-p) + \frac{pq}{(1-q)} \left[\sum_{n=1}^{\infty} n r^{n-1} - r^{1-1} \right]$$

$$= (1-p) + \frac{pq}{(1-q)} \left[\frac{1}{q^2} - 1 \right]$$

$$= (1-p) + \frac{pq}{q^2(1-q)} \left[1 - q^2 \right]$$

$$= (1-p) + \frac{pq}{q^2(1-q)} (1-q)(1+q) = (1-p) + \frac{p}{q}(1+q)$$

$$< \infty$$

** Compare with [Kleinrock] Theorem 2 on page 29, (2.45)

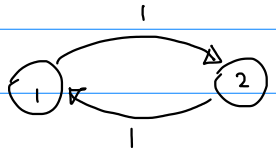
$p = q \Rightarrow (1-p) + 1 + p = 2$

↓ symmetric

$\frac{1}{M_1} = \frac{1}{2}$

$\frac{1}{M_2} = \frac{1}{2}$

Thus state (1) of above example is "recurrent nonnull"



$f_1^{(1)} = 0$ $f_1^{(3)} = 0$ $f_1^{(5)} = 0$

$f_1^{(2)} = 1$ $f_1^{(4)} = 0$ $f_1^{(6)} = 0$

.... all zero

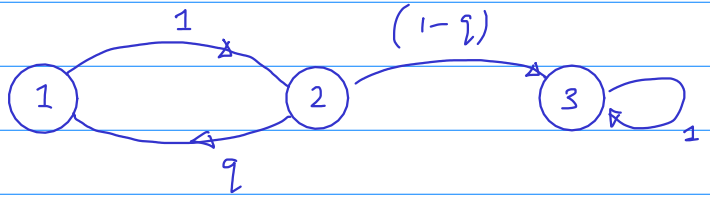
$$f_1 = \sum_{i=1}^{\infty} f_1^{(i)} = 1 \quad \longrightarrow \text{recurrent}$$

$$\sum_{i=1}^{\infty} i f_1^{(i)} = (2)\cdot(1) = 2 \quad \longrightarrow \text{"recurrent nonnull"}$$

Note $M_1 = 2$, $M_2 = 2$

$$\frac{1}{M_1} = \frac{1}{M_2} = \frac{1}{2}$$

compare with [Kleinrock] Theorem 2, page 29 equation (2.45)



$$f_1^{(1)} = 0$$
$$f_1^{(2)} = q$$
$$f_1^{(3)} = 0$$
$$f_1^{(4)} = 0$$

all zero

$$f_1^{(n)} = \begin{cases} q & n = 2 \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore F_1 = \sum_{n=1}^{\infty} f_1^{(n)} = q < 1$$

State 1 is a transient state

$$p_{11}^{(1)} = 0$$
$$p_{11}^{(2)} = q$$
$$p_{11}^{(3)} = 0$$
$$p_{11}^{(4)} = q^2$$
$$\vdots$$

$$\therefore p_{11}^{(n)} = \begin{cases} q^{n/2} & n \text{ even} \\ 0 & \text{otherwise} \end{cases}$$

$$\sum_{n=1}^{\infty} p_{11}^{(n)} = \sum_{n=1}^{\infty} q^n = q \sum_{n=1}^{\infty} q^{n-1} = q \frac{1}{1-q}$$

$$= \frac{q}{1-q} < \infty$$

compare with Ross Optimization Models book Proposition 4.1 on page 196

* m-step probability

$$p_{ij}^{(m)} = P\left[X_{n+m} = j \mid X_n = i \right]$$

$$= \sum_k P\left[X_{n+m} = j, X_{n+(m-1)} = k \mid X_n = i \right]$$

$$= \sum_k P\left[X_{n+m} = j \mid X_{n+(m-1)} = k, X_n = i \right] P\left[X_{n+(m-1)} = k \mid X_n = i \right]$$

$$= \sum_k \underbrace{P\left[X_{n+m} = j \mid X_{n+(m-1)} = k \right]}_{p_{kj}} p_{ik}^{(m-1)}$$

$$= \sum_k p_{ik}^{(m-1)} p_{kj}$$

$$P\left[X_2 = j\right] = \sum_k P\left[X_2 = j, X_1 = k\right]$$

↓ averaging out one {Random variable} to compute the marginal

$$P[X, Y \mid Z] = P[X \mid Y, Z] \cdot P[Y \mid Z]$$

$\forall y \quad \exists z \quad s.t. \quad y = z^3$

↓

Let y be arbitrary, Let $z = y^{1/3}$

$\therefore z^3 = (y^{1/3})^3 = y$ } above statement is true !

$P = \exists y \quad \forall z \quad y = z^3$

$\sim P = \forall y \quad \exists z \quad y \neq z^3$ → of course this is true !.

$\therefore$ P is FALSE. ! } above statement is false !